



# Beyond Grain Yield-Integrating Milling Quality and Nutritional Value into White Maize Breeding for Smallholder Systems in Zimbabwe: A Critical Review

Rachel Magaya<sup>1</sup>, Fortunate Makore<sup>2</sup>, Tinashe Zenda<sup>2</sup>,  
Cosmos Magorokosho<sup>3</sup>, Kenedy Simango<sup>4</sup>

10.18805/ag.RF-435

## ABSTRACT

In Zimbabwe, maize breeding has historically prioritized agronomic performances, particularly yield, over critical quality traits that determine consumer acceptance and nutritional outcomes. This focus has created a disconnect between officially released hybrids and the preferences of farmers, millers and consumers, potentially undermining both food security and public health in this maize-dependent economy. We critically synthesized literature through several academic databases to: (i) trace the evolution of breeding objectives in Zimbabwe; (ii) evaluate the status and adoption barriers of biofortification programs; (iii) analyze the socio-economic and cultural drivers of milling quality preferences and (iv) identify critical research gaps in evaluating new white maize hybrids. The review reveals that milling quality remains systematically under-evaluated despite its direct relationship to cultural acceptability (*sadza* texture) and economic value. Biofortification efforts for provitamin A, zinc and quality protein have generated nutritionally superior varieties, yet adoption remains constrained by persistent consumer preference for white maize, invisibility of nutritional traits, lack of market incentives and nutrient stability concerns. Critically, the nutritional content of new white maize hybrids is often assumed uniform without empirical verification, despite evidence that hybridization for agronomic traits can inadvertently alter grain composition. We argue for a dual strategy: continuing biofortification while instituting mandatory nutritional and milling quality screening for all new white maize hybrids during advanced on-farm trials. This integrated approach is essential to ensure new varieties meet the holistic needs of farmers, processors and consumers.

**Key words:** Biofortification, Food security, Maize, Milling quality, Nutritional value, Zimbabwe.

Maize (*Zea mays* L.), commonly known as corn, is one of the world's most widely grown cereals, ranked third in production after wheat and rice (Subba *et al.*, 2022). It is the cornerstone of food security in sub-Saharan Africa, providing 30-50% of daily caloric intake in Southern Africa (Galani *et al.*, 2022). In Zimbabwe, smallholder farmers produce 60-70% of the national crop (CLAFA-2, 2025), with the grain primarily consumed as *sadza*-a thick white porridge of profound cultural significance (Serna-Saldivar, 2016). Despite this centrality, a fundamental disconnect persists between how maize is bred and how it is ultimately used and valued.

Zimbabwe's maize breeding programs have historically prioritized agronomic traits: grain yield, drought tolerance, disease resistance and early maturity (Mukaro *et al.*, 2024; Nyakurwa *et al.*, 2017). While these objectives are justified given the imperative of food availability in a climate-vulnerable region, they have systematically neglected milling and nutritional quality. Milling quality encompasses grain characteristics affecting dry-milling efficiency and the organoleptic properties of *sadza*, directly influencing consumer preferences (Abdala *et al.*, 2018). Food security addresses not only calorie availability but also nutrient availability (Kujur *et al.*, 2025). Nutritional quality refers to protein quality and micronutrient content-

<sup>1</sup>Department of Horticulture, Faculty of Plant and Animal Sciences and Technology, Marondera University of Agricultural Sciences and Technology, P.O. Box 35, Marondera, Zimbabwe.

<sup>2</sup>Department of Marketing, Marondera University of Agricultural Sciences and Technology, P.O. Box 35, Marondera, Zimbabwe.

<sup>3</sup>Agricultural Research Council Complex, No. 51283 Hatcliffe Estate, Alpes Road, Harare, Zimbabwe.

<sup>4</sup>African Indigenous Vegetables Institute, Marondera University of Agricultural Sciences and Technology, P.O. Box 35, Marondera, Zimbabwe.

**Corresponding Author:** Rachel Magaya, Department of Horticulture, Faculty of Plant and Animal Sciences and Technology, Marondera University of Agricultural Sciences and Technology, P.O. Box 35, Marondera, Zimbabwe. Email: rmagaya@muast.ac.zw

**ORCID:** <https://orcid.org/0009-0006-3577-8666>, <https://orcid.org/0009-0001-5403-1756>, <https://orcid.org/0000-0002-7680-1520>, <https://orcid.org/0000-0003-3177-1791>, <https://orcid.org/0000-0001-6042-6028>

**How to cite this article:** Magaya, R., Makore, F., Zenda, T., Magorokosho, C. and Simango, K. (2026). Beyond Grain Yield-Integrating Milling Quality and Nutritional Value into White Maize Breeding for Smallholder Systems in Zimbabwe: A Critical Review. *Agricultural Reviews*. 1-11. doi: 10.18805/ag.RF-435.

**Submitted:** 03-04-2026 **Accepted:** 04-09-2026 **Online:** 29-09-2026

provitamin A, zinc and iron (Goredema-Matongera *et al.*, 2021). With maize dominating daily diets, its nutritional density carries serious public health implications, particularly in communities with limited dietary diversity (Kairiza *et al.*, 2020).

The recognition of conventional maize's nutritional deficiencies spurred biofortification efforts. Quality Protein Maize (QPM), developed using the *opaque-2* mutation, offers improved protein quality (Amegbor *et al.*, 2022). Provitamin A-enriched orange maize addresses vitamin A deficiency, while high-zinc maize is under development (Matongera *et al.*, 2023). Despite demonstrated nutritional efficacy, adoption in Zimbabwe remains minimal. Farmers overwhelmingly continue producing conventional white maize due to deep cultural preference for white grain (orange maize is often associated with animal feed); the "invisibility" of nutritional traits; absence of price premiums; seed system constraints; and concerns about nutrient stability during storage and processing (Nkhata *et al.*, 2024; Malézieux *et al.*, 2024).

While biofortification represents a long-term strategy, conventional white maize will dominate Zimbabwean diets for the foreseeable future. Yet the nutritional and milling quality of new white hybrids remains unexamined. Extensive hybridization and selection for yield and stress tolerance can inadvertently alter grain composition (Álvarez-Iglesias *et al.*, 2021). Traits such as endosperm hardness, protein content and micronutrient concentrations are genetically variable and can shift through breeding. Critically, Zimbabwe's variety release system does not require nutritional or milling quality assessments; a hybrid can be approved on agronomic performance alone.

This review synthesizes existing literature to: (i) trace the evolution of maize breeding objectives in Zimbabwe from the colonial era to present; (ii) evaluate biofortification programs (QPM, provitamin A, zinc), assessing achievements and adoption barriers; (iii) analyze the socio-economic and cultural drivers of milling quality preference among farmers, millers and consumers; and (iv) identify research gaps in the evaluation of new white maize hybrids and propose an integrated framework for future research that bridges the divide between agronomic performance and end-use quality.

## Conceptual framework: understanding maize quality beyond yield

### Defining milling quality

Milling quality refers to dry-milling efficiency and the characteristics of the resulting mealie-meal, determined by interacting grain physical and biochemical properties (Abdala *et al.*, 2018; Borrás *et al.*, 2022). Key determinants include endosperm hardness (vitreousness), where hard glossy endosperm yields efficient milling and acceptable *sadza*, while soft endosperm produces culturally unacceptable pasty results (Borrás *et al.*, 2022); kernel density and test weight, indicating hardness and milling yield; pericarp thickness, where thinner pericarp improves

bran separation and extraction rate; optimal moisture content of 12.5-15% (Suri and Tanumihardjo, 2016) and kernel size uniformity, reducing waste and improving consistency. In Zimbabwe, the ultimate test of a maize variety is *sadza*: firm, cohesive, non-sticky, with a grainy texture (Chimonyo *et al.*, 2019). Varieties failing this face consumer rejection regardless of yield potential.

### Defining nutritional quality

Maize is the source of the cheapest nutrients, though its nutritional quality is strengthened with a combination of other food materials (Yadav *et al.*, 2025). Maize protein (6-12%) is deficient in lysine and tryptophan, contributing to malnutrition in maize-dependent populations (Kaur *et al.*, 2020). Conventional white maize is also low in provitamin A, iron and zinc, fuelling hidden hunger; biofortification aims to raise these levels (Matongera *et al.*, 2023). Phytates reduce mineral bioavailability, while provitamin A degrades during storage and processing (Huey *et al.*, 2023).

### The interconnectedness of milling and nutritional quality

The dry-milling process removes the germ and bran, where most micronutrients concentrate (iron, zinc, lipids, fat-soluble vitamins) (Galani *et al.*, 2022), depleting the refined product nutritionally. Biofortification must therefore account for nutrient losses during milling (Palacios-Rojas *et al.*, 2020). Meaningful hybrid evaluation must simultaneously consider milling performance and nutritional outcomes within Zimbabwe's specific production and consumption context.

This framework guides the analysis that follows, emphasizing that meaningful hybrid evaluation must simultaneously consider milling performance and nutritional outcomes within Zimbabwe's specific context.

## Historical evolution of maize breeding in Zimbabwe: from yield-only to multi-trait approaches

### The colonial era and SR52 legacy

Commercial maize breeding in Southern Rhodesia began in the 1930s, following the development of maize hybrids in the United States (de Haas and Giller, 2016). The program, initially serving large-scale commercial farmers, produced SR52 in the 1960s-Africa's first single-cross hybrid for tropical environments. SR52 offered exceptional yield potential and standability, becoming the regional benchmark and establishing Zimbabwe as a maize improvement leader (Derera and Musimwa, 2015). However, SR52 was designed for high-input systems, ill-suited to resource-constrained smallholders relegated to marginal lands under colonial tenure. This created a bifurcated seed system: high-yielding hybrids for commercial farmers versus unimproved landraces for smallholders.

### Post-independence: expanding access

Following independence in 1980, the government prioritized expanding smallholder access to improved

varieties. The national breeding program (Department of Research and Specialist Services) shifted focus to varieties for low-input, heterogeneous smallholder conditions (Mukaro *et al.*, 2024). The 1980s-1990s saw releases of open-pollinated varieties and hybrids emphasizing yield stability, drought tolerance, disease resistance (MSV, GLS, TLB), lodging resistance and husk cover. These objectives boosted productivity but entrenched a yield-centric paradigm, with quality traits remaining secondary.

### The shift to participatory approaches

The late 1990s-2000s recognized that station-bred varieties often failed on smallholder farms due to environmental heterogeneity and farmer preference diversity. Participatory variety selection (PVS) emerged, engaging farmers as co-evaluators under their own management conditions (Begna, 2022). PVS studies in Zimbabwe (Magaisa *et al.*, 2022; Nyakurwa *et al.*, 2022) revealed that farmers' criteria diverged from breeders' priorities; they valued grain hardness, white color and taste alongside yield and stress tolerance. These findings provided early evidence that quality traits influence adoption yet have not been systematically integrated into breeding objectives.

### Current breeding objectives

Contemporary maize breeding in Zimbabwe, led by the Crop Breeding Institute in partnership with the International Maize and Wheat Improvement Center (CIMMYT) and the International Institute of Tropical Agriculture (IITA), has broadened beyond yield (Table 1). Key observations:

- Milling quality is addressed indirectly (e.g., "flint or semi-flint grain" as farmer preference), without direct targets for extraction rate, processing energy, or sadza sensory traits.
- Nutritional traits are pursued mainly in dedicated bio fortification programs; conventional white hybrids are not systematically screened for nutritional content.
- On-farm evaluation through PVS emphasizes agronomic performance and farmer rankings, rarely incorporating miller input or laboratory grain quality analysis (Mukaro *et al.*, 2024).

### Biofortification in Zimbabwe: progress, adoption barriers and lessons learned

#### Scientific basis and development

The 1960s discovery of the opaque-2 mutation increased endosperm lysine and tryptophan by 69% over conventional maize, dramatically improving protein quality (Kaur *et al.*, 2020). However, original mutants had soft, chalky endosperm, increased pest susceptibility and lower yield. Decades of CIMMYT breeding introgressed modifier genes to restore hard, vitreous endosperm while retaining the nutritional advantage, resulting in QPM that combines improved protein quality with agronomic performance comparable to conventional maize (Prasanna *et al.*, 2020).

### QPM in Zimbabwe: Releases and adoption

CIMMYT has been the primary QPM germplasm source, with CBI developing locally adapted varieties, releasing OPVs ZS 242 and ZS 243 and hybrids SC527 and SC643 (FAO, 2024; Setimela *et al.*, 2017). Studies confirm QPM diets, with ~30% more lysine and ~55% more tryptophan, improved children's growth outcomes (Kairiza *et al.*, 2020; Prakash *et al.*, 2017).

Despite proven efficacy, adoption remains limited and project-driven rather than commercially widespread (Maqbool *et al.*, 2021). Key barriers include:

- **Trait invisibility:** QPM is visually identical to conventional white maize, so farmers cannot perceive the nutritional advantage without price premiums or incentives.
- **Lack of market differentiation:** Grain aggregators and millers do not distinguish QPM, meaning nutritional benefits reach consumers but not farmers' incomes.
- **Yield perceptions:** Although modern QPM hybrids have narrowed yield gaps, persistent perceptions of yield disadvantage discourage adoption (Tandzi *et al.*, 2017).
- **Seed system constraints:** QPM seed production requires careful management to maintain the opaque-2 trait; contamination with conventional pollen can reduce nutritional quality.
- **GMO confusion:** Despite being conventionally bred, QPM is sometimes mistaken for GM maize, generating unwarranted scepticism.

### Critical evaluation

The QPM experience demonstrates that nutritional improvement alone is insufficient to drive adoption; traits must be bundled with other farmer-valued characteristics and institutional innovations (market linkages, quality assurance) are as important as genetic ones. Some economists question whether QPM investment represents the most cost-effective nutrition intervention, arguing that dietary diversification (legumes, animal-source foods) offers more holistic solutions (Pingali, 2015). This critique has merit but overlooks that for many resource-constrained households, maize will remain the dietary foundation for the foreseeable future; improving its nutritional quality remains a legitimate public health priority.

### Provitamin A maize

#### Scientific basis and development

Vitamin A deficiency (VAD) causes impaired immunity, increased mortality and preventable blindness, particularly among children and pregnant women in Zimbabwe (Goredema-Matongera *et al.*, 2021). Breeding programs, led globally by HarvestPlus and implemented in Zimbabwe through CBI and CIMMYT, have developed maize varieties with elevated provitamin A content (HarvestPlus, 2015). Targeted levels of ~7.5 ppm are sufficient to meet 25% of daily vitamin A requirements (Kairiza *et al.*, 2020). SC 642,

**Table 1:** Current breeding objectives in Zimbabwe's maize improvement programs.

| Primary challenge                                                                             | Key breeding objective                                              | Specific trait sought                                                                                                                                                                                                                                                                                                                         | Examples of varieties                                                                                                                   | Specific references                                                                                   |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Drought and terminal drought stress                                                           | -Develop drought tolerant (DT)<br>-Early-to-mid maturity varieties. | -ASI (Anthesis-Silking Interval):<br>Reduced gap between pollen shed and silk emergence.<br>-Root System: Deeper and more prolific roots for water scavenging.<br>-Stay-Green: Retains leaves green longer under stress.<br>-Early Maturity: Escapes terminal drought.                                                                        | ZM 309, ZM 421, SC 403, SC 529, PAN 3M-49, PAN 7M-81                                                                                    | 1. Lunduka <i>et al.</i> (2019).<br>2. Fisher <i>et al.</i> (2015).<br>3. Makore <i>et al.</i> (2021) |
| Low soil nitrogen (N) Fertility (Linked with smallholder systems where fertilizer use is low) | Develop nitrogen-use efficient (NUE) varieties.                     | -Efficient nitrogen uptake and utilization.<br>-Ability to maintain yield under low N conditions.                                                                                                                                                                                                                                             | ZM 309, ZM 521, ZM 621 (OPVs).<br>-Many DT hybrids are also selected for NUE.                                                           | 1. Makore <i>et al.</i> (2022)<br>2. Hamadziripi <i>et al.</i> (2025)                                 |
| Diseases and pests                                                                            | Develop disease-resistant varieties.                                | -Turicum leaf blight.<br>-Gray leaf spot.<br>-Maize streak virus.<br>-Common rust.<br>-Very short growing cycle (90-120 days).<br>-Stability in low and erratic rainfall environments.                                                                                                                                                        | -Modern hybrids are routinely rated for resistance (e.g., GLS and TLB ratings on seed packets)-SC 727<br>SC 301, SC 302, SC 303, ZM309. | 1. Seed Co International (2021)<br>2. Jaison <i>et al.</i> (2023).<br>3. Matova <i>et al.</i> (2023). |
| Expansion into marginal areas and climate change.                                             | Develop extra-early and ultra-early maturing varieties.             | -Stability in low and erratic rainfall environments.<br>-Heat tolerance during flowering.                                                                                                                                                                                                                                                     |                                                                                                                                         | 1. Das <i>et al.</i> (2025)<br>2. Mutenje (2016)<br>3. Mukaro <i>et al.</i> (2024)                    |
| Farmer and market preferences                                                                 | Improve agronomic and quality traits.                               | -Plant Stature: Semi-dwarf plants for standability.<br>-Grain Type: Primarily white, flint, or semi-flint grain.<br>-Cob Placement: Good husk cover.<br>-High Shelling Percentage.<br>For Commercial Farmers:<br>-High-yielding potential.<br>-High response to inputs.<br>For Smallholder Farmers:<br>-Yield stability and stress tolerance. | -Shift from tall OPVs (e.g., SR52) to shorter, sturdier modern hybrids.<br>-White grain                                                 | 1. Mukaro <i>et al.</i> (2024)<br>2. Fisher <i>et al.</i> (2015)                                      |
| Socio-economic diversity of farmers                                                           | Provide a portfolio of varieties for different farmer categories.   |                                                                                                                                                                                                                                                                                                                                               | -High-Input Hybrids: SC 727, PANNAR 7M-91.<br>-Medium-Input Hybrids: SC 403, PAN 3M-53.<br>-OPVs: ZM 421, ZM 625.                       | 1. Kassie <i>et al.</i> (2017)<br>2. Tarekne <i>et al.</i> (2023)                                     |

Table 1: Continue

Table 1: Continue

|                                               |                                                                           |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                          |                                                                                                                                                                                                                  |
|-----------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 'Hidden hunger' or micronutrient malnutrition | Develop biofortified maize varieties with enhanced micronutrient content. | -Affordability.<br>-Provitamin A (PVA): Increased levels of carotenoids (beta-carotene), which the body converts to Vitamin A.<br>- Zinc: Higher zinc content in the grain.<br>-Iron: Higher grain iron content<br>-Amino acids: Improved protein quality, such as higher lysine and tryptophan content (e.g., Quality Protein Maize - QPM). | -Orange Maize (Provitamin A): SC 643. (Other PVA Varieties under demonstration and promotion by the Government and NGOs<br>-Quality Protein Maize (QPM). | 1. Nyakurwa <i>et al.</i> (2022)<br>2. HarvestPlus (2018)<br>3. Goredema Matongera <i>et al.</i> (2021)<br>4. Setimela <i>et al.</i> (2017)<br>5. Amegbor <i>et al.</i> (2022)<br>6. Tandzi <i>et al.</i> (2017) |
|-----------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

released by Seed Co, was Zimbabwe's first provitamin A hybrid, identifiable by its orange grain color (HarvestPlus, 2015).

Adoption barriers: The color challenge

Zimbabwe's deep cultural preference for white maize remains the greatest adoption barrier, as orange or yellow maize is associated with animal feed or inferior quality. Though blind taste tests showed consumer acceptance increases with nutritional information. Hence real-world adoption requires sustained behaviour change communication, which remains inconsistently implemented.

Nutrient stability concerns

A second significant concern is the stability of provitamin A carotenoids during storage and processing. Beta-carotene is susceptible to oxidative degradation during storage and processing under high temperatures and light exposure, with substantial provitamin A losses documented post-harvest (Syawalluddin *et al.*, 2024; Hamieh *et al.*, 2023). A variety may test high at harvest but deliver diminished nutritional impact upon consumption. Breeding programs must therefore select for both high initial Provitamin A content and carotenoid stability under typical post-harvest conditions.

Cost-effectiveness debate

Provitamin A maize has faced critiques regarding cost-effectiveness. Malézieux *et al.* (2024) argue that over-reliance on a single biofortified staple may divert resources from dietary diversification, which addresses multiple deficiencies simultaneously. Other vitamin A-rich foods, such as orange-fleshed sweet potato, mangoes, dark leafy greens and animal-source foods, are available and should be promoted. Biofortification should therefore complement, not substitute for, dietary diversification efforts.

Zinc biofortification

Current status

Zinc deficiency impairs immune function, growth and development, contributing significantly to the global burden of disease (Lowe *et al.*, 2024). Biofortification of maize with zinc is technically more challenging than provitamin A enhancement because zinc content is not visually discernible; laboratory analysis is required to verify trait expression. The HarvestPlus program has established a target of 38 ppm zinc in maize grain (HarvestPlus, 2015).

In Zimbabwe, high-zinc breeding lines are in the early stages of testing and adaptation by CBI (Matongera *et al.*, 2023). The trait is often being 'stacked' with other biofortified traits, developing varieties that combine provitamin A, QPM and high zinc in a single genetic background (Matongera *et al.*, 2023).

Methodological challenges

Zinc biofortification faces key challenges:

- **Soil dependency:** Genetic potential for grain zinc is limited by soil zinc availability. High-zinc varieties may underperform

on zinc-deficient soils common in smallholder systems, requiring agronomic biofortification (zinc-enriched fertilizers) to ensure consistency (Manzeke-Kangara *et al.*, 2021).

- **Health impact measurement:** Zinc deficiency is hard to diagnose at population level due to non-specific symptoms; linking health improvements to biofortified maize requires costly, rigorous studies with advanced biomarkers (Khan *et al.*, 2022).

- **Marketing invisibility:** Like QPM, zinc-enriched maize lacks visual or taste differentiation. Creating market incentives would require certification and labeling systems that do not yet exist.

## Milling quality: The neglected trait

### Grain physical traits and milling outcomes

Milling quality encompasses grain characteristics determining dry-milling efficiency and the organoleptic properties of the final product. In Zimbabwe, where 95% of maize is consumed as refined white mealie-meal (Melesse *et al.*, 2021), the ultimate expression of milling quality is sadza quality, characterized by firmness, cohesiveness, non-stickiness, grainy texture and white color.

These sensory properties are directly influenced by grain physical traits. Hard endosperm genotypes produce larger, coarser particles during milling, rehydrating into firm, non-sticky sadza. Soft endosperm genotypes yield more fine particles, producing a pasty, sticky consistency considered low-quality (Borrás *et al.*, 2022; Chimonyo *et al.*, 2019). Key traits include endosperm hardness (vitreousness), affecting grit yield, fines and energy use; measured via the Stenvert test, NIRS, or floatation methods; test weight, indicating kernel soundness; kernel uniformity and shape; pericarp thickness and optimal moisture content (12.5-15%) for dry milling (Walker *et al.*, 2023).

### Economic and cultural drivers

Milling quality has direct economic consequences. Farmers benefit from price premiums and reject varieties with poor sadza quality. Millers prioritize extraction rate, as hard endosperm reduces costs and increases premium output (Walker *et al.*, 2023). Consumers influence demand through brand loyalty in urban markets and satisfaction with home-processed grain in rural areas.

Despite these clear drivers, milling quality is not systematically evaluated in Zimbabwe's variety testing protocols. The phenotyping bottleneck, requiring larger grain samples and specialized equipment, is a partial explanation. However, with NIRS offering rapid, non-destructive prediction of grain quality traits, this bottleneck is now surmountable (Alamu *et al.*, 2021; DeSalvio *et al.*, 2024).

### The nutritional trade-off in milling

A critical and often overlooked aspect is the relationship between milling and nutritional outcomes. Producing refined

white mealie-meal, the culturally preferred product, requires removing the germ and bran fractions, precisely where micronutrients (iron, zinc, lipids, B vitamins) are concentrated (Galani *et al.*, 2022; Suri and Tanumihardjo, 2016). Milling removes approximately 50-80% of iron and zinc.

For provitamin A maize, this trade-off is particularly concerning, as carotenoids concentrate in the germ. If milling removes the germ, much of the nutritional investment is lost. Palacios-Rojas *et al.* (2020) emphasize that biofortification breeding must consider nutrient retention after milling and cooking, varieties concentrating provitamin A in the endosperm, rather than the germ, may be preferable.

This milling-nutrition trade-off has profound implications: biofortification alone is insufficient if nutrients are removed during processing. Milling and nutritional quality must be evaluated together, with processing studies assessing nutrient retention under typical milling and cooking conditions as part of variety evaluation protocols.

## The research gap

The preceding analysis reveals a fundamental gap in Zimbabwe's maize research and development system: new white maize hybrids are released without empirical data on their milling quality or nutritional content. This gap persists despite evidence that hybridization and selection for agronomic traits can alter grain composition. Studies have documented significant genetic variation in protein content, amino acid profiles, micronutrient concentrations and endosperm hardness among maize genotypes (Álvarez-Iglesias *et al.*, 2021; Goredema-Matongera *et al.*, 2023; Iljkić *et al.*, 2025). Breeding for yield, drought tolerance, or disease resistance may have correlated effects on these quality traits, either positive or negative. The assumption that white maize hybrids are nutritionally equivalent or adequate is empirically unsupported. Several lines of evidence challenge this assumption:

### Genetic variation

Maize exhibits substantial genetic diversity for kernel composition. Langyan *et al.* (2021) found significant variation in protein, oil, starch and micronutrient content among maize inbred lines, indicating that breeding choices can shift composition.

### Genotype × environment interaction

Nutrient content is not fixed; it varies with growing conditions. Greveniotis *et al.* (2023) and Matongera *et al.* (2023) demonstrated that grain quality traits are influenced by both genotype and environment and that genotype rankings for quality can change across environments. On-farm evaluation under smallholder conditions is essential to understand real-world nutritional outcomes.

### Breeding trade-offs

Selection for high yield can inadvertently reduce nutrient concentration through 'dilution effects' if yield increases

are not accompanied by proportional increases in nutrient uptake and partitioning. This has been documented for minerals in various crops, though evidence in maize is mixed.

### **The distinctiveness of hybrids**

Each new hybrid represents a novel combination of alleles. Assuming that its nutritional profile mirrors that of previous hybrids is scientifically unwarranted. Empirical measurement is required. This gap has two critical implications for research and policy.

### **Establishment of nutritional baselines**

Variety release should document nutritional composition, including protein quality, zinc, iron and provitamin A. This would establish baselines, identify exceptional varieties, detect nutritional degradation and inform stakeholders. Nutritional quality assessments should be standard for all advanced hybrids, not limited to biofortified varieties. This ensures the entire maize germplasm base contributes to nutrition security.

### **Identification of incremental improvement opportunities**

On-farm trials identify white maize varieties with superior nutrition under realistic conditions, accounting for genotype x environment interactions. Evaluating nutritional traits across locations helps identify consistent performers, understand environmental determinants and engage farmers. This approach moves beyond the white versus biofortified binary, elevating nutrition across all maize while respecting consumer preferences.

### **Methodological innovations: integrating quality traits into breeding programs**

This section reviews innovations enabling integrated quality assessment and proposes a framework for implementation in Zimbabwe.

#### **High-throughput phenotyping technologies**

Near-Infrared Spectroscopy (NIRS) substantially reduces the phenotyping bottleneck. It rapidly predicts protein, amino acids, micronutrients and milling traits from a single sample (Alamu *et al.*, 2021). While requiring an initial investment, a centralized NIRS facility in Zimbabwe could enable systematic quality screening across breeding programs, universities and seed companies.

#### **Participatory approaches to quality evaluation**

Participatory variety selection complements laboratory analysis by capturing culturally specific sensory properties. Structured sadza tastings with farmers, millers and consumers generate acceptability data. PVS studies reveal farmers' trait trade-offs, including culinary quality. Women's perspectives on processing, cooking and sadza quality are essential and must be deliberately included (Chopera *et al.*, 2022).

#### **An integrated framework for quality evaluation**

Drawing on the above, we propose an integrated framework for evaluating white maize hybrids that combines laboratory phenotyping with participatory approaches:

#### **Stage 1: Early-generation screening (breeder nurseries).**

NIRS analysis of advanced lines for protein, oil, starch and hardness enables selection for acceptable quality before yield testing intensifies and identifies lines with exceptional nutritional properties.

#### **Stage 2: Multi-location trials (research stations)**

Continue NIRS analysis across environments to assess stability. Add targeted biochemical analyses (lysine/tryptophan, zinc, provitamin A) for promising materials. Measure test weight, thousand-kernel weight and other physical traits.

#### **Stage 3: On-farm participatory evaluation**

Continue NIRS across environments; Add targeted biochemistry. Measure physical traits.

#### **Stage 4: Miller engagement**

Evaluate with millers; measure extraction rate, energy; assess meal color and particle size.

#### **Stage 5: Integrated analysis and recommendation**

Synthesize data; identify varieties optimizing across criteria; develop recommendations highlighting quality.

This framework embeds quality evaluation throughout the breeding pipeline, generating data to inform breeding decisions, guide recommendations and provide transparent information to farmers and consumers.

## **FUTURE PERSPECTIVES**

### **Summary of key findings**

Through this thorough review, we are able to provide an overview of the current state of maize breeding, biofortification and milling quality research within Zimbabwe, which is indicative of a systematic disconnect between breeding interests and the needs of the maize value chain. The long-standing focus on yield and stress tolerance, while justified based on food security considerations, has resulted in a significant disconnect where official hybrids may be adequate in agronomic considerations but not necessarily adequate in regards to the requirements of farmers, millers and consumers. This review makes clear that milling quality, including factors like grain hardness and endosperm vitreousness, as well as those affecting cultural acceptability of the sadza, is systematically underestimated despite being directly related to economic value and consumer preference. On the other hand, biofortification projects have yielded improved nutritional varieties effective at addressing protein and micronutrient deficiencies; however, implementation has been limited by the continued cultural preference for white maize, lack of market visibility of the nutritionally-enhanced traits, inadequate seed systems and genuine worries about nutritional stability. Perhaps most importantly, however, it becomes clear that there is a core weakness within Zimbabwe's maize research process, in that the nutritional and milling qualities of the new white maize hybrids being developed is taken for granted and does not

require any testing at all before their release. Such an assumption is scientifically unsound, since there is ample scientific evidence available that the process of hybridization and selective breeding for agronomic qualities may have an unintended impact on the composition of grains, as well as strong interactions between genotypes and environments affecting quality traits. It is argued that there should be two strategies pursued simultaneously, biofortification and quality screening of all new white maize hybrids.

### Implications for stakeholders

For breeders, quality traits must be prioritized equally with yield and stress tolerance, supported by NIRS screening of all advanced materials and participatory approaches, including women in sensory evaluation. Biofortification should continue alongside nutritional screening of all white hybrids. Policymakers should amend variety release protocols to require documentation of milling quality and nutritional composition, establish minimum standards and support centralized phenotyping facilities. Seed companies should highlight quality attributes in marketing and engage millers to inform product development. Researchers need multi-location on-farm trials measuring agronomic and quality traits, studies on nutrient retention under storage and processing and genetic analysis of milling quality. Extension services should incorporate quality attributes into messaging, provide nutrition education on processing methods that preserve nutrients and support farmer-managed quality evaluation.

### CONCLUSION

Maize is not simply a source of calories in Zimbabwe; it is the foundation of food security, a vector of cultural identity and a determinant of public health. Breeding programs that focus exclusively on yield, however successful, address only part of what maize means to the millions who depend on it. The argument of this review is not that yield is unimportant; it is essential. Rather, we argue that yield alone is insufficient. Milling quality determines whether grain becomes acceptable food; nutritional quality determines whether that food nourishes its consumers. The way forward requires a paradigm shift: from a narrow focus on production to a holistic vision of food systems. This means breeding maize that not only yields well but also mills efficiently, cooks into culturally acceptable sadza and provides meaningful nutrition to those who consume it. It means evaluating new white maize hybrids not just for their performance in research station trials, but for their performance in the hands of farmers, in the mills of processors and on the plates of families. The tools to achieve this vision exist. High-throughput phenotyping makes quality trait evaluation feasible at scale. Participatory approaches ensure that evaluation reflects real-world preferences. Breeding programs that integrate these tools can develop maize hybrids that serve the whole food system, not just the production subsector.

For Zimbabwe and for the millions of maize-dependent households across sub-Saharan Africa, the stakes could not be higher. The next generation of maize varieties will shape food security and public health for decades to come. Ensuring that they are not only high-yielding but also high-quality is not a technical nicety; it is a moral imperative.

### ACKNOWLEDGEMENT

The present study was supported by InnovAfricgenetics (Pvt) Ltd and Marondera University of Agricultural Sciences and Technology, which provided time and facilities. Additionally, the first author, Rachel Magaya, is thankful to InnovAfricgenetics and the Agricultural Research Council for supporting this Research.

### Disclaimers

The views and conclusions expressed in this article are solely those of the authors and do not necessarily represent the views of their affiliated institutions. The authors are responsible for the accuracy and completeness of the information provided, but do not accept any liability for any direct or indirect losses resulting from the use of this content.

### Informed consent

This is a review article based on published literature and does not involve any human or animal participants; therefore, informed consent is not applicable.

### Conflict of interest

The authors declare that there are no conflicts of interest regarding the publication of this article. No funding or sponsorship influenced the design of the study, data collection, analysis, decision to publish, or preparation of the manuscript.

### REFERENCES

- Abdala, L.J., Gambin, B.L. and Borrás, L. (2018). Sowing date and maize grain quality for dry milling. *European Journal of Agronomy*. **92**: 1-8.
- Alamu, E.O., Nuwamanya, E., Cornet, D., Meghar, K., Adesokan, M., Tran, T. and Davrieux, F. (2021). Near-infrared spectroscopy applications for high-throughput phenotyping for cassava and yam: A review. *International Journal of Food Science and Technology*. **56(3)**: 1491-1501.
- Álvarez-Iglesias, L., Malvar, R.A., Garzón, R., Rosell, C.M. and Revilla, P. (2021). Nutritional value of whole maize kernels from diverse endosperm types and effects on rheological quality. *Agronomy*. **11(12)**: 2509.
- Amegbor, I., van Biljon, A., Shargie, N., Tareknege, A. and Labuschagne, M. (2022). Identifying quality protein maize inbred lines for improved nutritional value of maize in Southern Africa. *Foods*. **11(7)**: 898.
- Begna, T. (2022). Importance of participatory variety selection and participatory plant breeding in variety development and adoption. *Journal of Plant Sciences and Crop Protection*. **5(2)**: 104.

- Borrás, L., Caballero-Rothar, N.N., Saenz, E., Segui, M. and Gerde, J.A. (2022). Challenges and opportunities of hard endosperm food grade maize sourced from South America to Europe. *European Journal of Agronomy*. **140**: 126596.
- Chimonyo, V.G.P., Mutengwa, C.S., Chidzuza, C. and Tandzi, L.N. (2019). Participatory variety selection of maize genotypes in the Eastern Cape Province of South Africa. *South African Journal of Agricultural Extension*. **47(1)**: 103-117.
- Chopera, P., Zimunya, P.R., Mugariri, F.M. et al. (2022). Facilitators and barriers to the consumption of traditional foods among adults in Zimbabwe. *J. Ethn. Food*. **9**: 5. <https://doi.org/10.1186/s42779-022-00121-y>.
- CLAFA-2. (2025). Second Round of Crops, Livestock and Fisheries Assessment Report. Ministry of Lands, Agriculture, Fisheries, Water and Rural Development, Zimbabwe.
- Das, B., Mutiga, S.K., Odiyo, O., Madahana, S., Milic, D., Sinyinda, L., Mwansa, K. et al. (2025) Costing of the breeding operations for the national maize programs in Eastern and Southern Africa. *Front. Sustain. Food Syst*. **9**: 1545600. doi: 10.3389/fsufs.2025.1545600.
- de Haas, M. and Giller, K.E. (2016). Pathways to African food security: Challenges, threats and opportunities towards 2050. Routledge.
- Derera, J. and Musimwa, T.R. (2015). Why is SR52 such a great maize hybrid? I. Heterosis and generation mean analysis. *Euphytica*. **205(1)**: 121-135.
- DeSalvio, A.J., Adak, A., Murray, S.C., Jarquin, D., Winans, N.D., Crozier, D. and Rooney, W.L. (2024). Near-infrared reflectance spectroscopy phenomic prediction can perform similarly to genomic prediction of maize agronomic traits across environments. *The Plant Genome*. **17(2)**: e20454.
- FAO. (2024). The State of Food Security and Nutrition in the World. Financing to end hunger, food insecurity and malnutrition in all its forms. In The State of Food Security and Nutrition in the World 2024. FAO; IFAD; UNICEF; WFP; WHO; <https://doi.org/10.4060/cd1254en>.
- Fisher, M., Abate, T., Lunduka, R.W., Asnake, W., Alemayehu, Y. and Madulu, R.B. (2015). Drought-tolerant maize for farmer adaptation to drought in sub-Saharan Africa: Determinants of adoption in eastern and southern Africa. *Climatic Change*. **133(2)**: 283-299.
- Galani, Y.J.H., Ligowe, I.S., Kieffer, M., Kamalongo, D., Kambwiri, A.M., Kuwali, P., Thierfelder, C., Dougill, A.J., Gong, Y.Y. and Orfila, C. (2022). Conservation agriculture affects grain and nutrient yields of maize (*Zea Mays* L.) and can impact food and nutrition security in Sub-Saharan Africa. *Frontiers in Nutrition*. **8**: 804663. <https://doi.org/10.3389/fnut.2021.804663>.
- Galani, Y.J.H., Orfila, C. and Gong, Y.Y. (2022). A review of micro nutrient deficiencies and analysis of maize contribution to nutrient requirements of women and children in Eastern and Southern Africa. *Critical Reviews in Food Science and Nutrition*. **62(6)**: 1568-1591.
- Goredema-Matongera, N., Ndhlela, T., Magorokosho, C., Kamutando, C.N., van Biljon, A. and Labuschagne, M. (2021). Multinutrient biofortification of maize (*Zea mays* L.) in Africa: Current status, opportunities and limitations. *Nutrients*. **13(3)**: 1039.
- Goredema-Matongera, N., Ndhlela, T., van Biljon, A., Kamutando, C.N., Cairns, J.E., Baudron, F. and Labuschagne, M. (2023). Genetic variation of zinc and iron concentration in normal, provitamin A and quality protein maize under stress and non-stress conditions. *Plants*. **12(2)**: 270.
- Greveniotis, V., Bouloumpasi, E., Zotis, S., Korkovelos, A., Kantas, D. and Ipsilandis, C.G. (2023). Stability dynamics of main qualitative traits in maize cultivations across diverse environments regarding soil characteristics and climate. *Agriculture*. **13(5)**: 1033.
- Hamadziripi, E.T., Collinson, S., Voss, R.C., Baudron, F., Labuschagne, M.T., Franke, A.C. and Cairns, J.E. (2025). Validating a novel genetic technology for hybrid maize seed production under management practices associated with resource-poor farmers in Zimbabwe. *Plants, People, Planet*. **7(3)**: 801-815.
- Hamieh, B., Borel, P., Raouche, S., Bruzzese, L., Adjriou, N., Halimi, C., Marconot, G., Gillet, G., Rostain, J.C., Guieu, R. and Desmarchelier, C. (2023). Post-harvest atmospheric pressure and composition modify the concentration and bioaccessibility of  $\alpha$ - and  $\beta$ -Carotene in carrots and sweet potatoes. *Foods*. **12(23)**: 4262. <https://doi.org/10.3390/foods12234262>.
- HarvestPlus. (2015). Going Global 2014 Annual Report. HarvestPlus.
- HarvestPlus. (2018). Crop Product Profile: Vitamin A Maize Zimbabwe. HarvestPlus.
- Huey, S.L., Konieczynski, E.M., Mehta, N.H., Krisher, J.T., Bhargava, A., Friesen, V.M. and Mehta, S. (2023). A systematic review of the impacts of post-harvest handling on provitamin A, iron and zinc retention in seven biofortified crops. *Nature Food*. **4(11)**: 978-985.
- Iljić, D., Rastija, M., Šimić, D., Lončarić, Z., Drenjančević, L., Zebec, V. and Varga, I. (2025). Effects of extreme combined abiotic stress on yield and quality of maize hybrids. *Agronomy*. **15(6)**: 1440.
- Jaison, E., Mawira, D. and Sangombe, N. (2023). Evaluating performance of CIMMYT maize inbred lines against maize streak virus (MSV), grey leaf spot (GLS), Rust (PS), Turcicum (ET) and yield potential in Zimbabwe. *Journal of Emerging Technologies and Innovative Research*. **10(5)**: 234-245.
- Kairiza, T., Kembo, G., Pallegedara, A. and MacHeka, L. (2020). The impact of food fortification on stunting in Zimbabwe: Does gender of the household head matter? *Nutrition Journal*. **19(1)**: 1-11.
- Kassie, G.T., Abdulai, A., Greene, W.H., Shiferaw, B., Abate, T., Tarekegne, A. and Sutcliffe, C. (2017). Modelling preference and willingness to pay for drought tolerance (DT) in maize in rural Zimbabwe. *World Development*. **94**: 465-477. <https://doi.org/10.1016/j.worlddev.2017.02.008>.
- Kaur, R., Kaur, G., Vikal, Y., Gill, G.K., Sharma, S., Singh, J., Dhariwal, G.K., Gulati, A., Kaur, A., Kumar, A. and Chawla, J.S. (2020). Genetic enhancement of essential amino acids for nutritional enrichment of maize protein quality through marker-assisted selection. *Physiology and Molecular Biology of Plants: An International Journal of Functional Plant Biology*. **26(11)**: 2243-2254. <https://doi.org/10.1007/s12298-020-00897-w>.

- Khan, S.T., Malik, A., Alwarthan, A. and Shaik, M.R. (2022). The enormity of the zinc deficiency problem and available solutions; An overview. *Arabian Journal of Chemistry*. **15(3)**: 103668.
- Kujur, A.N., Barla, S. and Upasani, R.R. (2025). Climate-smart agriculture: A pathway to food security and climate resilience: A review. *Agricultural Reviews*. **47(3)**: 446-450. doi: 10.18805/ag.R-2777.
- Langyan, S., Dar, Z.A., Chaudhary, D. P., Shekhar, J.C., Herlambang, S., El Enshasy, H. and Rakshit, S. (2021). Analysis of nutritional quality attributes and their inter-relationship in maize inbred lines for sustainable livelihood. *Sustainability*. **13(11)**: 6137.
- Lowe, N.M., Hall, A.G., Broadley, M.R., Foley, J., Boy, E. and Bhutta, Z.A. (2024). Preventing and controlling zinc deficiency across the life course: A call to action. *Advances in Nutrition*. **15(3)**: 100181.
- Lunduka, R.W., Mateva, K.I. and Magorokosho, C. (2019). Impact of adoption of drought-tolerant maize varieties on total maize production in South-Eastern Zimbabwe. *Climate and Development*. **11(8)**: 700-709.
- Magaisa, A., Manjeru, P., Kamutando, C.N. and Moyo, M.P. (2022). Participatory variety selection and stability of agronomic performance of advanced sorghum lines in Zimbabwe. *Journal of Crop Improvement*. **36(3)**: 440-460.
- Makore, F. (2021). Molecular and genetic analysis of maize (*Zea mays* L.) Inbred Lines under Abiotic Stress Environments. PhD Thesis, University of Zimbabwe.
- Makore, F., Kamutando, C.N., Nyoni, R.S., Dari, S., Gasura, E. and Mazarura, U. (2021). Understanding heat and drought stress adaptation mechanisms in maize on the molecular level. *Cereal Research Communications*. **49(4)**: 521-527.
- Makore, F., Magorokosho, C., Dari, S., Gasura, E., Mazarura, U. and Kamutando, C.N. (2022). Genetic potential of new maize inbred lines in single-cross hybrid combinations under low-nitrogen stress and optimal conditions. *Agronomy*. **12(9)**: 2205.
- Malézieux, E., Verger, E.O., Avallone, S., Alpha, A., Ngigi, P.B., Lourme-Ruiz, A., Bazile, D., Bricas, N., Ehret, I., Martin-Prevel, Y. and Amiot, M.J. (2024). Biofortification versus diversification to fight micronutrient deficiencies: An *Interdisciplinary Review*. *Food Security*. **16(1)**: 261-275. <https://doi.org/10.1007/s12571-023-01422-z>.
- Manzeke-Kangara, M.G., Joy, E.J.M., Mtambanengwe, F., Chopera, P., Watts, M.J., Broadley, M.R. and Mapfumo, P. (2021). Good soil management can reduce dietary zinc deficiency in Zimbabwe. *CABI Agriculture and Bioscience*. **2(1)**: 1-12.
- Maqbool, M.A., Beshir Issa, A.R. and Khokhar, E.S. (2021). Quality protein maize (QPM): Importance, genetics, timeline of different events, breeding strategies and varietal adoption. *Plant Breeding*. **140(3)**: 375-399.
- Matongera, N., Ndhlela, T., van Biljon, A. and Labuschagne, M. (2023). Predicting zinc-enhanced maize hybrid performance under stress conditions. *Food and Energy Security*. **12(4)**: e479.
- Matova, P.M., Kamutando, C.N., Warburton, M.L., Williams, W.P., Magorokosho, C., Shimelis, H. and Gowda, M. (2023). New techniques for breeding maize (*Zea mays*) varieties with fall armyworm resistance and market-preferred traits for sub-Saharan Africa. *Plant Breeding*. **142(1)**: 1-11.
- Melesse, M.B., Tirra, A.N., Ojiewo, C.O. and Hauser, M. (2021). Understanding farmers' trait preferences for dual-purpose crops to improve mixed crop-livestock systems in Zimbabwe. *Sustainability*. **13(10)**: 5678.
- Mukaro, R., Chaingeni, D., Sneller, C., Cairns, J.E., Musundire, L., Prasanna, B.M. and Kutwayo, D. (2024). Genetic trends in the Zimbabwe's national maize breeding program over two decades. *Frontiers in Plant Science*. **15**: 1391926.
- Mutenje, M. (2016). A climate risk profile of maize value chain farming system in Malawi, Zambia and Zimbabwe: Vulnerability assessment report. CIMMYT.
- Nkhata, S.G., Watson, F., Chimbaza, M., Namaumbo, S., Kammwamba, K., Chisapo, G., Ngoma, T.N., Chilemba, M. and Matumba, L. (2024). Farmers preferentially allocate more land to cultivation of conventional white maize compared to weevil-resistant biofortified orange maize. *Sustainability*. **16(19)**: 8600. <https://doi.org/10.3390/su16198600>.
- Nyakurwa, C.S., Gasura, E. and Mabasa, S. (2017). Potential for quality protein maize for reducing protein-energy undernutrition in maize-dependent Sub-Saharan African countries: A review. *African Crop Science Journal*. **25(4)**: 521. <https://doi.org/10.4314/acsj.v25i4.9>
- Nyakurwa, C.S., Gasura, E., Mabasa, S., Rugare, J.T. and Setimela, P.S. (2022). On-farm agronomic performance and farmer preference of quality protein maize grown under conservation agriculture in Southern Africa: A case for Zimbabwe. *Journal of Agricultural Science*. **160(3-4)**: 168-184.
- Palacios-Rojas, N., McCulley, L., Kaeppler, M., Titcomb, T.J., Gunaratna, N.S., Lopez-Ridaura, S. and Tanumihardjo, S.A. (2020). Mining maize diversity and improving its nutritional aspects within agro-food systems. *Comprehensive Reviews in Food Science and Food Safety*. **19(4)**: 1809-1834.
- Pingali, P. (2015). Agricultural policy and nutrition outcomes-Getting beyond the preoccupation with staple grains. *Food Security*. **7(3)**: 583-591.
- Prakash, J., Lata, S., Sharma, A., Vekariya, R.D., Sharma, J.K. and Sood, A. (2017). Evaluation of quality protein maize (*Zea mays* L.) hybrids for quantitative and quality traits. *International Journal of Current Microbiology and Applied Sciences*. **6(4)**: 1851-1859. <https://doi.org/10.20546/ijcmas.2017.604.220>.
- Prasanna, B.M., Palacios-Rojas, N., Hossain, F., Muthusamy, V., Menkir, A., Dhliwayo, T. and Fan, X. (2020). Molecular breeding for nutritionally enriched maize: Status and prospects. *Frontiers in Genetics*. **10**: 1392.
- Seed Co International. (2021). Seed Co 2021 Annual Report. Seed Co Limited.
- Serna-Saldivar, S.O. (2016). Maize: Foods from Maize. In the Reference Module in Food Science. Elsevier.
- Setimela, P.S., Gasura, E. and Tarekegne, A.T. (2017). Evaluation of grain yield and related agronomic traits of quality protein maize hybrids in Southern Africa. *Euphytica*. **213(12)**: 1-17.

- Subba, V., Nath, A., Kundagrami, S. and Ghosh, A. (2022). Study of combining ability and heterosis in quality protein maize using line x tester mating design. *Agricultural Science Digest*. **42(2)**: 159-164. doi: 10.18805/ag.D-5460.
- Suri, D.J. and Tanumihardjo, S.A. (2016). Effects of different processing methods on the micronutrient and phytochemical contents of maize: From A to Z. *Comprehensive Reviews in Food Science and Food Safety*. **15(5)**: 912-926.
- Syawaluddin, N.S., Abdul Rahman, H., Lim, S.J., Wan Mustapha, W.A., Mohd Razali, N.S., Kasim, K.F. *et al.* (2024). Lycopene and  $\beta$ -carotene thermal degradation kinetics and colour-antioxidant changes in gac (*Momordica cochinchinensis*) fruit aril paste. *Int J Food Sci Technol*. **59**: 7808-7817. <https://doi.org/10.1111/ijfs.17005>.
- Tandzi, L.N., Mutengwa, C.S., Ngonkeu, E.L.M., Woïn, N. and Gracen, V. (2017). Breeding for quality protein maize (QPM) varieties: A review. *Agronomy*. **7(4)**: 80.
- Tarekegne, A., Wegary, D., Cairns, J.E., Zaman-Allah, M., Beyene, Y., Negera, D. *et al.* (2023). Genetic gains in early maturing maize hybrids developed by the International maize and wheat improvement center in Southern Africa during 2000-2018. *Frontiers in Plant Science*. **14**. <https://doi.org/10.3389/fpls.2023.1321308>.
- Walker, C.K., Assadzadeh, S., Wallace, A.J., Delahunty, A.J., Clancy, A.B., McDonald, L.S. and Panozzo, J.F. (2023). Technologies and data analytics to manage grain quality on farm: A review. *Agronomy*. **13(4)**: 1129.
- Yadav, A., Singh, U. and Chaudhary, G. (2025). Development of maize-based composite flour *Cheela* and its effect on nutritional, sensory characteristics and quality parameters. *Asian Journal of Dairy and Food Research*. **45(4)**: 651-656. doi: 10.18805/ajdfr.DR-2224.